# SKaMP. Tests

The goal of this report is to provide information on the performed testing of data acquisition, data pre-processing, batch processing and streaming processing.

## 1 Test Scenarios

| Test objective | Steps | Expected Result | Actual Result |
|---|---|---|---|
| Verify data incoming from stream API | 1. Start the server using start_containers.bat; 2. Navigate to http://localhost:5000 | Incoming data is available on /data/0 | Passed. The screenshot is provided in Figure 3 |
| Verify correct setup of the stream and data preprocessing functions | Run 'pytest' from the root folder | Data stream is configured as expected; Incoming data is not null; Returned status code - 200. Preprocessing utils return transformed data as expected | Passed. The screenshot is provided in Figure 12 |
| Verify the correct setup of Nifi - HDFS/Kafka flow | Run the containers - follow steps in README.md | Data flows from streamin API to Kafka topics and Hive tables | Passed. The screenshot is provided in Figure 4, 5, 6, 7 |
| Verify the correct setup of batch processing | Run the containers - follow steps in README.md | Views are available in the Cassandra tables | Passed. The screenshot is provided in Figure 8, 9, 10 |
| Verify the correct setup of streaming processing | Run the containers - follow steps in README.md | Views are available in the Cassandra tables | Passed. The screenshot is provided in Figure 11 |
| Verify that Superset correctly connects to Cassandra through TrinoDB and displays data in dashboards | Run the necessary containers (start_containers) in 'scripts' folder and observe the charts in Superset | Data is visible in Superset charts | PASSED. Screenshots are visible in Figure x and x |

Table 1: Test scenarios

| Test objective | Steps | Expected Result | Actual Result |
|---|---|---|---|
| Verify correct data preprocessing of dataset 1 | Run 'pytest' from the root folder | Feature 'type' is correctly transformed into numeric value (5 cases); Feature 'isMerchant' is correctly prepared (2 cases) | PASSED. The screenshot is provided in Figure 12 |
| Verify correct data preprocessing of dataset 2 | Run 'pytest' from the root folder | Numeric boolean values are transformed to int from float (4 cases) | PASSED. The screenshot is provided in Figure 12 |
| Verify correct data preprocessing of dataset 3 | Run 'pytest' from the root folder | Feature 'entry_mode' is correctly transformed into numeric value (4 cases); Unnecessary features are omitted. | PASSED. The screenshot is provided in Figure 12 |
| Verify correct data preprocessing of dataset 4 | Run 'pytest' from the root folder | Features 'Amount', 'Class' are renamed to 'amount' and 'isFraud'; Extra features are removed | PASSED. The screenshot is provided in Figure 12 |

Table 2: Data pre-processing tests

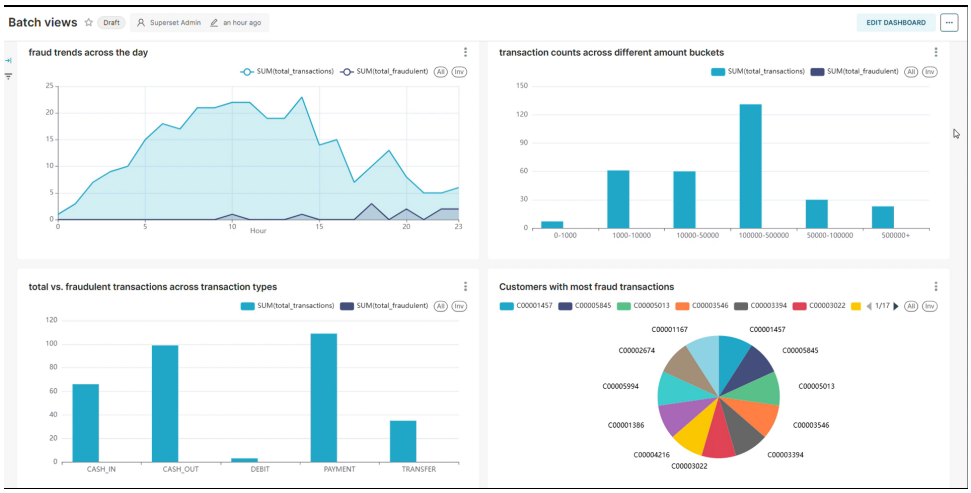



Figure 1: Batch Dashboard

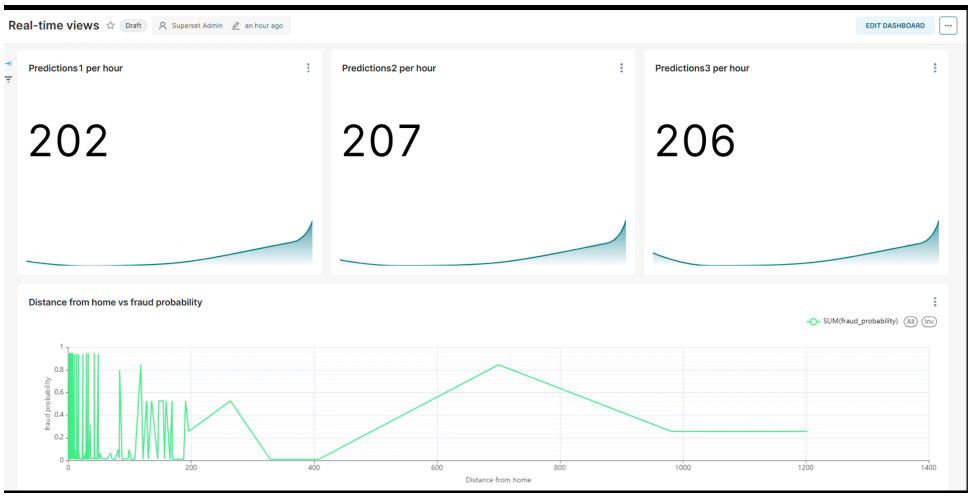



Figure 2: Speed Dashboard

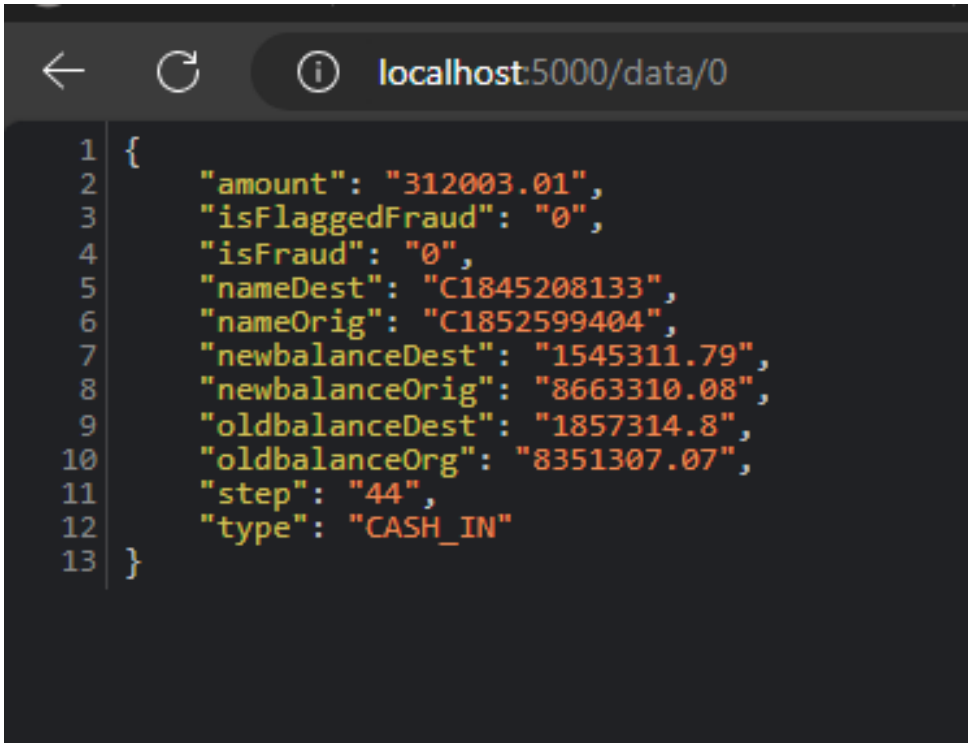



Figure 3: Data incoming via the stream




Figure 4: Kafka Dataset1

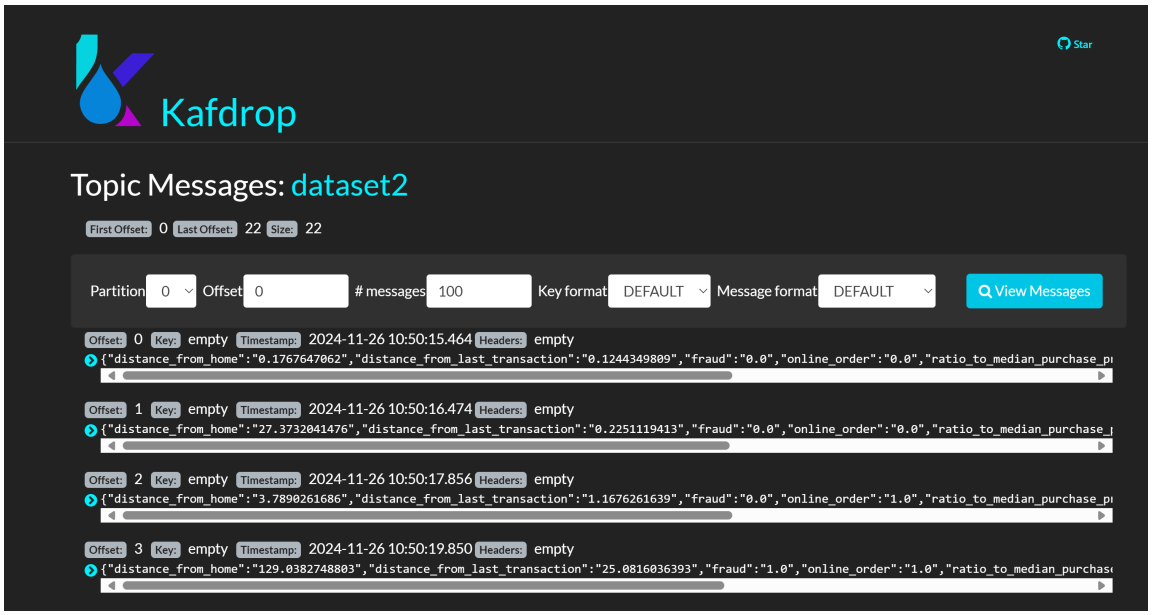



Figure 5: Kafka Dataset2

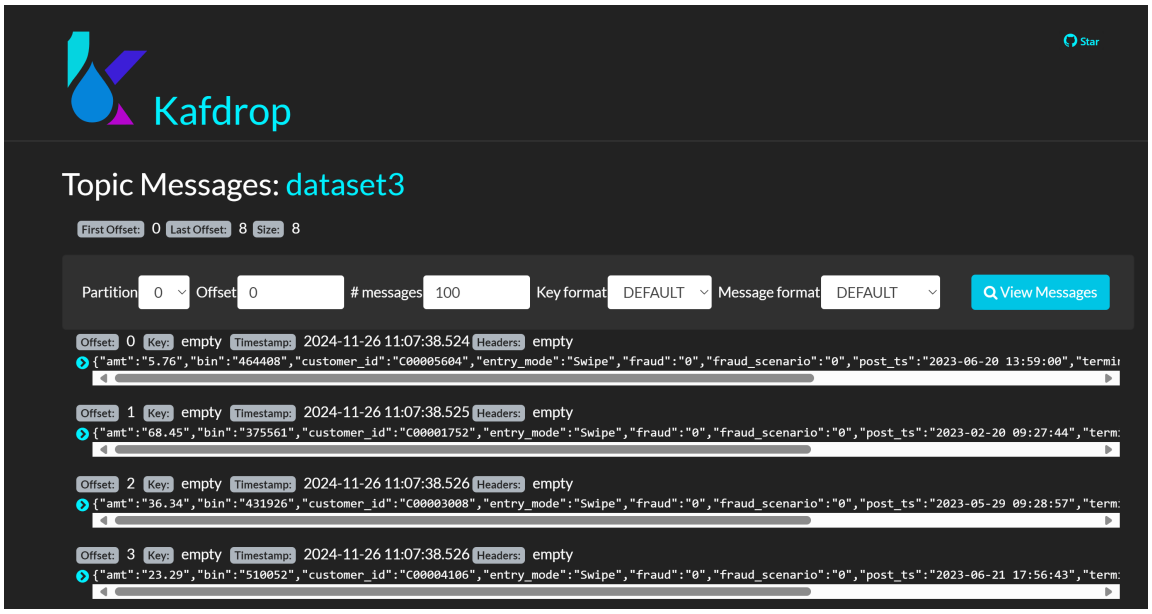



Figure 6: Kafka Dataset3

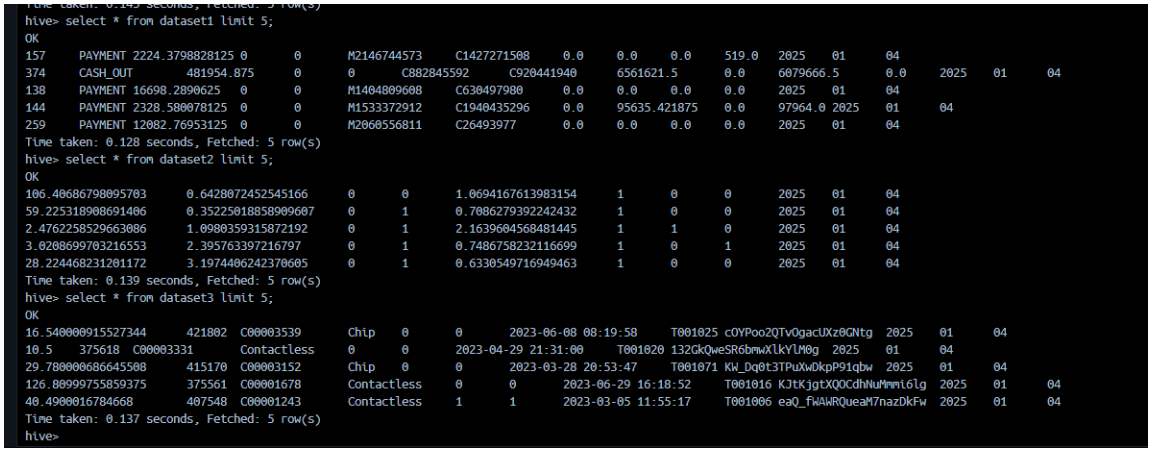



Figure 7: Hive data

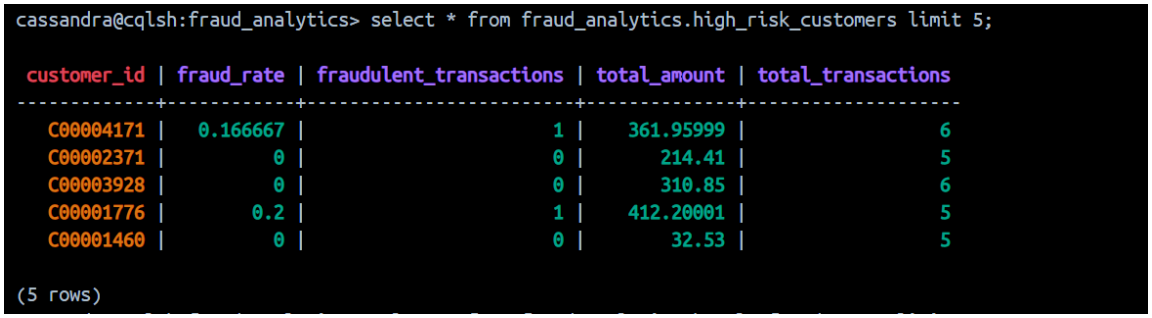



Figure 8: Cassandra batch processing views

```
cassandra@cqlsh:fraud_analytics> select * from fraud_analytics.hourly_fraud_stats limit 5;

 hour | avg_amount | total_fraudulent | total_transactions
------+------------+------------------+--------------------
   23 |   59.18323 |                6 |                 31
    5 |   54.17043 |                1 |                117
   10 |   55.43596 |                4 |                230
   16 |     55.149 |                6 |                190
   13 |   56.83678 |                3 |                242

(5 rows)
```

Figure 9: Cassandra batch processing views

```
cassandra@cqlsh:fraud_analytics> select * from fraud_analytics.fraud_by_transaction_type limit 5;

 type     | avg_amount   | fraud_rate | total_fraudulent | total_transactions
----------+--------------+------------+------------------+--------------------
 TRANSFER |  9.3788e+05 |   0.007463 |                2 |                268
 CASH_OUT |  1.7427e+05 |   0.001847 |                2 |               1083
    DEBIT |  5886.82766 |          0 |                0 |                 25
  CASH_IN |  1.7194e+05 |          0 |                0 |                708
  PAYMENT | 12990.00553 |          0 |                0 |               1165

(5 rows)
```

Figure 10: Cassandra batch processing views

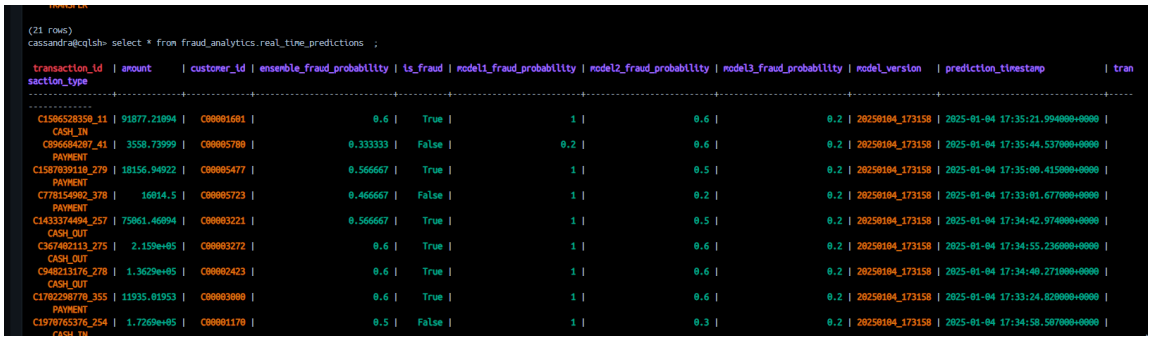



Figure 11: Cassandra stream processing views

```
============================================= test session starts =============================================
platform win32 -- Python 3.13.0, pytest-8.3.3, pluggy-1.5.0 -- C:\ProgramFiles\Anaconda3\envs\bigdata13\python.exe
cachedir: .pytest_cache
rootdir: C:\home\WUT\Semester_3\BigData\Big-Data-Analytics
collected 12 items

services/streaming_simulation/test_streaming_simulation.py::StreamingSimulationTestCase::test_data_stream PASSED [  8%]
tests/data_utils/test_utils.py::test_preprocess_1_payment PASSED                                            [ 16%]
tests/data_utils/test_utils.py::test_preprocess_1_cash_in PASSED                                            [ 25%]
tests/data_utils/test_utils.py::test_preprocess_1_cash_out PASSED                                           [ 33%]
tests/data_utils/test_utils.py::test_preprocess_1_debit PASSED                                              [ 41%]
tests/data_utils/test_utils.py::test_preprocess_1_unknown PASSED                                            [ 50%]
tests/data_utils/test_utils.py::test_preprocess_row_2 PASSED                                                [ 58%]
tests/data_utils/test_utils.py::test_preprocess_3_contactless PASSED                                        [ 66%]
tests/data_utils/test_utils.py::test_preprocess_3_chip PASSED                                               [ 75%]
tests/data_utils/test_utils.py::test_preprocess_3_swipe PASSED                                              [ 83%]
tests/data_utils/test_utils.py::test_preprocess_3_unknown PASSED                                            [ 91%]
tests/data_utils/test_utils.py::test_preprocess_row_4 PASSED                                                [100%]

============================================= 12 passed in 0.57s ==============================================
```

Figure 12: Unit testing result

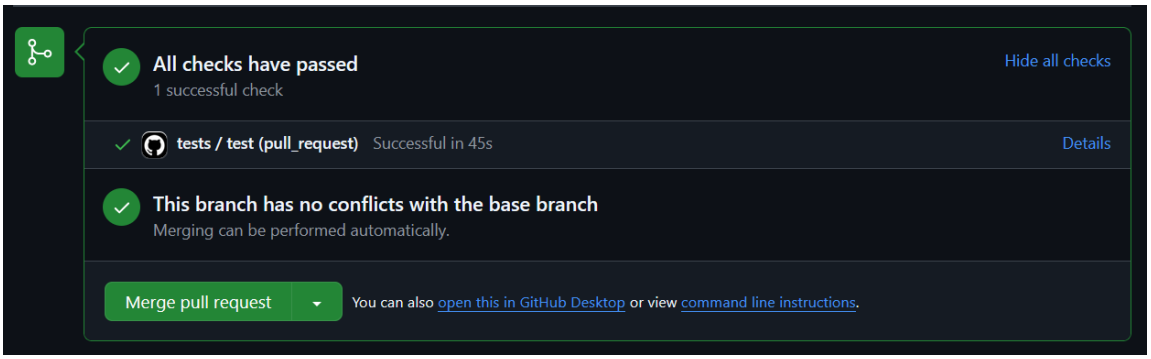



Figure 13: GitHub checks before merge